

JOONWON LEE

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SUMMARY

Statistics Ph.D. candidate and computational statistician specializing in scalable likelihood inference, stochastic-process modeling, numerical optimization, and statistical diagnostics for dependent data. Builds reproducible Python/PyTorch research systems, integrates C++/pybind11 for performance-critical computation, and evaluates model fit, uncertainty, and misspecification through simulation and empirical diagnostics.

EDUCATION

Ph.D. in Statistics, Rutgers University, Piscataway, NJ

Expected May 2027 | GPA: 3.8/4.0

M.S. in Statistics, University of Minnesota, Minneapolis, MN

Sep 2021 | GPA: 3.7/4.0

TECHNICAL SKILLS

Programming and Research Systems: Python (NumPy, Pandas, SciPy, PyTorch, LightGBM, scikit-learn), C++/pybind11 integration, R, SQL, Linux, Git, AWS EC2, HPC/SLURM; CPU/GPU computing, backtesting, and restartable research pipelines

Quantitative Methods: Stochastic-process and time-series modeling, Monte Carlo simulation, MLE/GLS, numerical optimization, spectral and autocorrelation analysis, hypothesis testing, model diagnostics and validation, backtesting, regression, and machine learning

QUANTITATIVE INDUSTRY EXPERIENCE

JPMorgan Chase - Summer Quantitative Analytics Associate, Model Risk Governance and Review

Summer 2026

- Developed a sequential risk-monitoring framework for binary indicators, using Brownian-motion approximation, dynamic-programming probability propagation, and simulation to quantify stopping-time uncertainty and early-decision tradeoffs.
- Specified and validated a Sequential Probability Ratio Test with explicit hypotheses and likelihood-ratio boundaries; evaluated boundary-crossing probabilities, stopping behavior, and Type I/II error.

Travelers - Data Science Leadership Program, Business Insurance Pricing Team

Summer 2025

- Developed an end-to-end LightGBM pricing pipeline across 2.46M+ records, including data preparation, feature processing, GLM benchmarking, model selection, validation, and performance diagnostics on AWS EC2; presented findings as pricing and risk-segmentation recommendations to business stakeholders.

QUANTITATIVE RESEARCH AND TRADING

IMC Prosperity Algorithmic Trading Competition - Top 2.39% of Participants

Apr 2025

- Built and backtested an inventory-aware market-making strategy around empirically calibrated reference values; tracked position-limit saturation over a rolling 10-step window and triggered more aggressive inventory-reducing quotes to manage persistent exposure.

Ph.D. Dissertation Research - Scalable Stochastic Modeling and Statistical Diagnostics

Sep 2024 - Present

- **Scalable Inference:** Developed an advection-aware Vecchia likelihood approximation for nonseparable space-time covariance structures, using ordered local conditional models for likelihood-based parameter estimation and uncertainty quantification with up to 145,008 observations per day.
- **Stochastic Diagnostics:** Developed statistical diagnostics for dependent stochastic processes that localize where a fitted model fails across scale and frequency, distinguishing persistent dependence from high-frequency noise, missingness, acquisition artifacts, and covariance misspecification.
- **Research Systems:** Built reusable Python/PyTorch workflows for data-quality filtering, time-dependent alignment, optimization, simulation, and restartable CPU/GPU and HPC execution; integrated performance-critical C++ ordering routines through pybind11.

ADDITIONAL

Coursework: Machine Learning, Linear Algebra, Probability Theory, Stochastic Processes, Statistical Computing, Advanced Theory of Statistics I-II, Data Structures and Algorithms, Microeconomics, Econometrics

Languages: Korean (Native); English (Fluent); Chinese (Basic)